
KTU
NOTES

ected so as to minimize the total
ile satisfying the load demand &
ational constraints.

ation of 2 main problems

Problem

Unit Commitment / Pre dispatch problem

on-line economic dispatch/Cost Minimization (satisfying certain Constraints)

to select optimally out of the
nerating sources to operate, to
pected load .

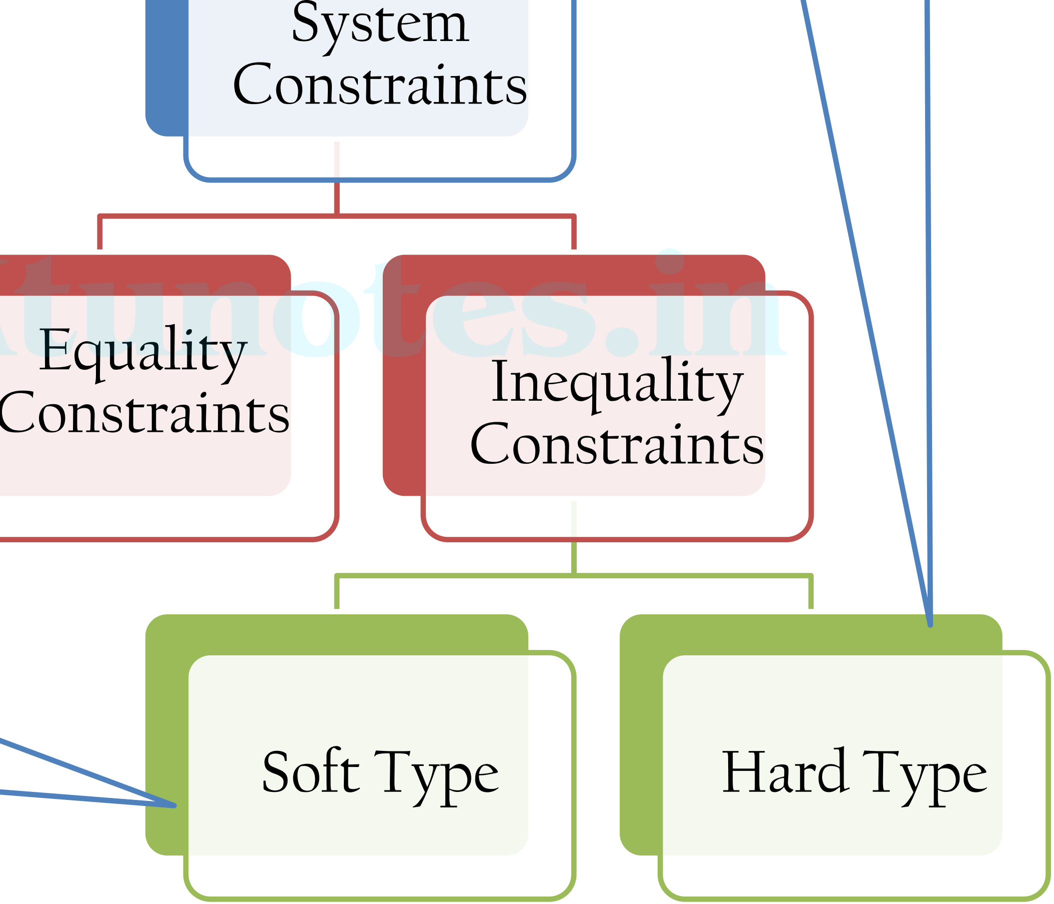
ecified margin of operating
a specified period of time.

to distribute the load among the
nits actually paralleled with the
ch manner as to minimize the total
lying the minute-to-minute
s of the system.

with minimum fuel consumption.

e one which is optimal in the sense minimum cost of generation.

generation is a function of the neration of the sources which can vithin certain constraints, the cost of ill depend upon the system constraint ar load demand.


System Constraints
Equality Constraints
Inequality Constraints
Soft Type
Hard Type

$$\sum_{q=1}^{n} \{e_p(e_q G_{pq} + f_q B_{pq}) + f_p(f_q G_{pq} - e_q B_{pq})\}$$

$$\sum_{q=1}^{n} \{f_p(e_q G_{pq} + f_q B_{pq}) - e_p(f_p G_{pq} - e_q B_{pq})\}$$

$1, 2, \ldots, n$

ot exceed a pre specified value Cp because of the
se conditions,

$$P_p^2 + Q_p^2 \leq C_p^2$$

output of a generator for optimum operation
n is less than a pre specified value Pmin the
ut on the bus bar because it is not possible to
low value of power from that unit.

generator powers Pp cannot be outside the
by the inequality

$$P_{p\,min} \le P_p \le P_{p\,max}$$

nd minimum reactive power generation of a ited.

um reactive power is limited because of of the rotor and minimum is limited because y limit of the machine.

nerator reactive power Qp cannot be outside ed by the inequality, i.e.

$$Q_{p\,min} \leq Q_p \leq Q_{p\,max}$$

l that the voltage magnitudes and phase
s nodes should vary within certain limits.

agnitudes should vary within certain limits
vise most of the equipment's connected to
not operate satisfactorily or additional use
gulating devices will make the system

$$_{min}| \le |V_p| \le |V_{p\,max}|$$

$$\delta_{p\,min} \le \delta_p \le \delta_{p\,max}$$

outages of one or more alternators on the

cted load on the system.

ation should be such that in addition to
ad demand and losses a minimum spare
ould be available i.e.,

$$G \geq P_p + P_{so}$$

e total generation and Pso is some pre
r. A well planned system is one in which
ity Pso is minimum.

e.,

$$0 \leq t \leq 1.0$$

winding transformer if tappings are provided on the

$$0 \leq t \leq n$$

of transformation. Phase shift limits of the phase shifting

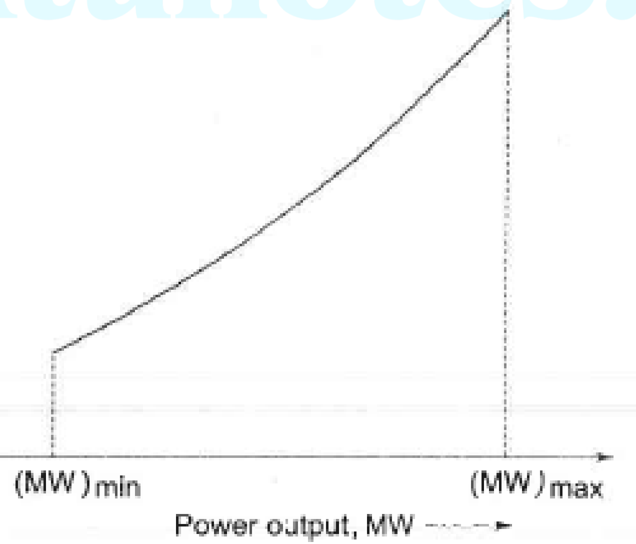
$$\theta_{p\,min} \leq \theta_p \leq \theta_{p\,max}$$

e and reactive power through the transmission line l by the thermal capability of the circuit and is

$$C_p \le C_{p\,max}$$

the maximum loading capacity of the pth line.

omponent of generator operating
uel input/hour, mainly for thermal
station.



Input-output curve of a generating unit

$$= \frac{1}{2} a_i P_{Gi}^2 + b_i P_{Gi} + d_i \ Rs/hr$$

uffix i stands for the unit number.

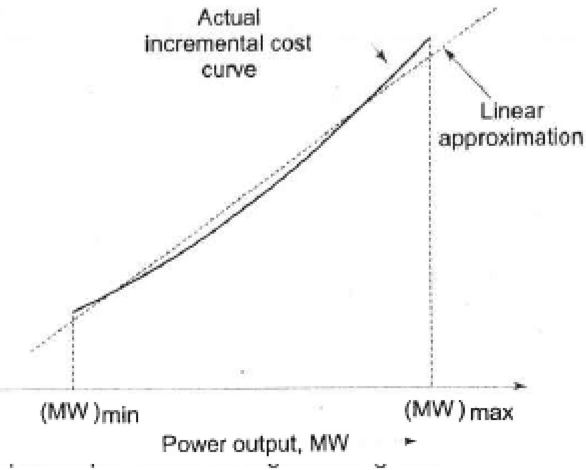
f the cost curve $\frac{dC_i}{dP_{Gi}}$ is called the

Fuel cost(IC)

r units of Rupees/Mega Watt

Whr).

$_i P_{Gi} + b_i$



$$= \sum_{i=1}^{n} P_i$$

input to the system.

ut to nth unit.

ad demand.

ion of nth unit.

$$F_T + \lambda\left(P_D - \sum_{i=1}^{n} P_i\right)$$

he Lagrangian multiplier

ng F with respect to the generation

ing to zero gives the condition for

ration of the system.

$$F_1 + F_2 + F_3 + \ldots\ldots + F_i + \ldots + F_n$$

$$= \lambda$$

re the condition for

peration is

$$= \ldots = \frac{dF_i}{dP_i} = \ldots = \frac{dF_n}{dP_n} = \lambda$$

$$P_L = \sum_{i=1}^{n} P_i$$

the total system loss which is
be a function of generation and the
ave their usual significance.

$$+\lambda\left(P_D + P_L - \sum_{i=1}^{n} P_i\right)$$

ne Lagrangian multiplier or the
cost of received power in

ng *F with respect to the generation*
ting to zero gives the condition for
ration of the system.

$$\partial P_i \quad)$$

$$- \frac{\partial P_L}{\partial P_i})$$

$$= F_1 + F_2 + F_3 + \ldots\ldots + F_i + \ldots. + F_n$$

$$(1 - \frac{\partial P_L}{\partial P_i})$$

$m \dfrac{\quad}{(1-\dfrac{\partial P_L}{\partial P_i})}$ *is known as*

or at i^{th} *unit* (L_i)

the condition for

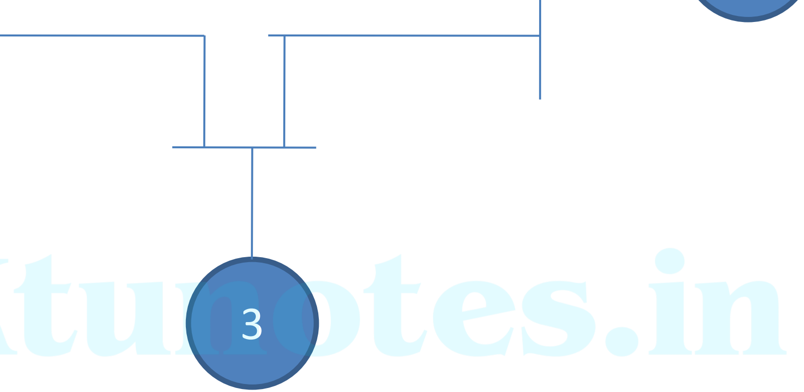
ration is

$$L_2 = \ldots = \frac{dF_i}{dP_i} L_i = \ldots = \frac{dF_n}{dP_n} L_n = \lambda$$

n loss at plant i

$$\sum_{m}\sum_{n} P_m B_{mn} P_n$$

nd Pn are the source loadings,
nsmission loss coefficients



Loss

$$ {}_{1} + B_{22}P_{G2}^2 + B_{33}P_{G3}^2 + 2B_{12}P_{G1}P_{G2} + 2B_{23}P_{G2}P_{G3} $$

$$ {}_{1}P_{G3}P_{G1} $$

ver system analysis-
merical Problems

rnator is expressed as $0.12\ P^2 + 20P+40$ ₹/hr.
rnator= 200MW.
C when the alternator is loaded at 75% of its
e IC and FC when the alternator is fully loaded?

$^2 + 20P+40$

$P + 20$

oad → $0.24*150 + 20 = 56$ Rs/Mw-hr

$150^2)+ 20(150)+40$

Rs/Hour

d IC→ $0.24(200) + 20 = 68$ Rs/Mw-Hr

$F= 0.12(200)^2 +20(200) +40$

$= 8840$ Rs/Hr

l inputs per hour of plants 1 and 2 are given as

$$F_1 = 0.2\, P_1^2 + 40\, P_1 + 120 \text{ Rs. per hr}$$

$$F_2 = 0.25\, P_2^{\,2} + 30\, P_2 + 150 \text{ Rs. per hr}$$

c operating schedule and the corresponding cost of generation if loading on each unit is 100 MW and 25 MW, the demand is osses are neglected. If the load is equally shared by both the tained by loading the units as per equal incremental production c operation of the units

$$\frac{dF_1}{dP_1} = \frac{dF_2}{dP_2}$$

$$0.4P_1 + 40 = 0.5P_2 + 30$$

$$P_1 + P_2 = 180$$

equations gives

$$P_1 = 88.89 \text{ MW} \quad \text{and} \quad P_2 = 91.11 \text{ MW}$$

ration = $F_1 + F_2$

$$F_1 = 0.2\,P_1^2 + 40\,P_1 + 120 = \text{Rs. } 5255.88/\text{hr.}$$

$$F_2 = 0.25\,P_2^2 + 30\,P_2 + 150 = \text{Rs. } 4958.55/\text{hr.}$$

Total cost = Rs. 10214.43/hr.

each unit is 90 MW, the cost of generation will be

$$F_1 = \text{Rs. } 5340/\text{hr.}$$

$$F_2 = \text{Rs. } 4875/\text{hr.}$$

Total cost = Rs. 10215/hr.

Rs. 0.57/hr.

$$F_1 = (8P_1 + 0.024\,P_1^{\,2} + 80)10^6$$

$$F_2 = (6P_2 + 0.04\,P_2^{\,2} + 120)10^6$$

d minimum loads on the units are 100 MW and 10 MW respectively.
cost of generation when the following load (Fig. 2..) is supplied.
er million Btu.

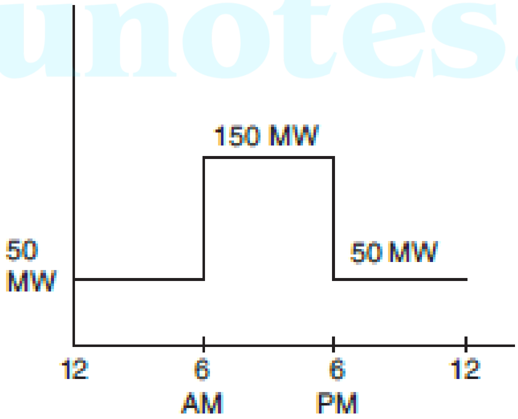


Fig 2

rom the fuel input characteristics

$$\frac{dF_1}{dP_1} = 0.048\, P_1 + 8$$

$$\frac{dF_2}{dP_2} = 0.08\, P_2 + 6$$

d is 50 MW, for economic loading the conditions are

$$\frac{dF_1}{dP_1} = \frac{dF_2}{dP_2}$$

$$P_1 + P_2 = 50$$

equations,

$$P_1 = 15.625 \text{ MW and } P_2 = 34.375 \text{ MW}$$

$$F_1 = 210.868 \text{ million Btu per hr.}$$

$$F_2 = 373.5 \text{ million Btu per hr.}$$

d is 150 MW, the equations are

$$0.048\,P_1 + 8 = 0.08P_2 + 6$$

$$P_1 + P_2 = 150$$

$$P_1 = 78.126 \text{ MW} \quad \text{and} \quad P_2 = 71.874 \text{ MW}$$

$$F_1 = 851.496 \text{ million Btu/hr.}$$

$$F_2 = 757.87 \text{ million Btu/hr.}$$

al cost = Rs. (210.868 + 373.5 + 851.496 + 757.87) × 12 × 2

= Rs. 52649.61 **Ans.**

to the load, a loss of 15.625 MW is incurred. Determine the generation emand if the cost of received power is Rs. 24/MWhr. Solve the problem ations and the penalty factor method approach. The incremental plants are

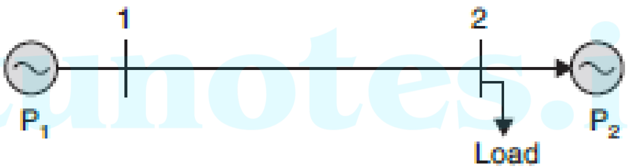


Fig 1

$$\frac{dF_1}{dP_1} = 0.025\,P_1 + 15$$

$$\frac{dF_2}{dP_2} = 0.05\,P_2 + 20$$

$$15.625 = B_{11} \times 125$$

$$B_{11} = 0.001$$

$dP_1 \qquad dP_2$

$For\ second\ unit, Loss = 0$

$(\because No\ transmission\ of\ power, load\ concentrated\ at\ unit\ 2)$

$\therefore B_{12} = B_{21} = B_{22} = 0$

$Total\ loss, P_L = B_{11}P_1^2 + 2B_{12}P_2 + B_{22}P_2^2 = B_{11}P_1^2$

<u>*For Plant 2*</u>

$$\frac{dF_2}{dP_2} L_2 = \lambda$$

$\lambda = Cost\ Received\ by\ the\ polant = 24\ Rs / MWhr$

$$\frac{dF_2}{dP_2} = 0.05P_2 + 20$$

$$Penalty\ factor, L_2 = \frac{1}{(1 - \frac{\partial P_L}{\partial P_2})} = 1, \frac{\partial P_L}{\partial P_2} = 0$$

$$\frac{dF_2}{dP_2} = \lambda \Rightarrow 0.05P_2 + 20 = 24$$

From this

$P_2 = 80\ MW$

$$\frac{1}{1 - 0.002P_1} = 24$$

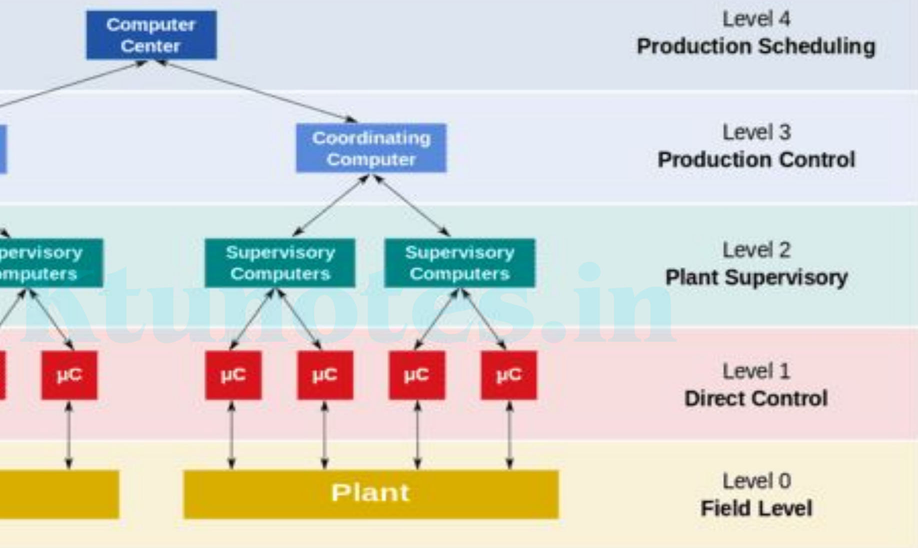
$$\frac{025P_1 + 15}{- 0.002P_1} = 24$$

$$P_1 = 123.28 \text{ MW}$$

is zero, $\therefore L_2$ = unity, *i.e.*, the incremental cost of received power

of production.

$P_2 + 20 = 24$ or $P_2 = 80$ MW



neral term for a high-level of overall control of many individual
rol loops. It gives the operations supervisor an overview of the
ntegration of operation between low-level controllers.

omic load dispatch is

ns

of economic load dispatch is

2 P_{G2}

quations

$P_{G2} = 107.5$ MW

ctively. Thus, the minimum cost dispatch will be

(1)

f economic load dispatch is

$$I_{c1} = I_{c2} \Rightarrow \frac{dC_1}{dPG_1} = \frac{dC_2}{dPG_2}$$

$$C_1 = PG_1 + 0.055P^2G_1$$

$$C_2 = 3PG_2 + 0.03P^2G_2$$

$$\frac{dC_1}{dPG_1} = 1 + 0.11PG_1$$

$$\frac{dC_2}{dPG_2} = 3 + 0.06PG2$$

$$\Rightarrow 1 + 0.11PG_1 = 3 + 0.06PG_2$$

$$0.11PG_1 - 0.06PG_2 = 2.....(2)$$

equations

$_2 = 150$ MW

o meet 300 MW of load, P_1 and P_2 respectively, are

at

200 MW

n condition of economic load dispatch is

$= b + 4cP_2$

P_2

e, $P_1 = 200$ MW and $P_2 = 100$ MW

$$incremental\ cost\ I_C = 0.012P + 8$$

$$\frac{dPL}{dP} = 0.2$$

$$\lambda = 25$$

$$We\ know\ that,$$

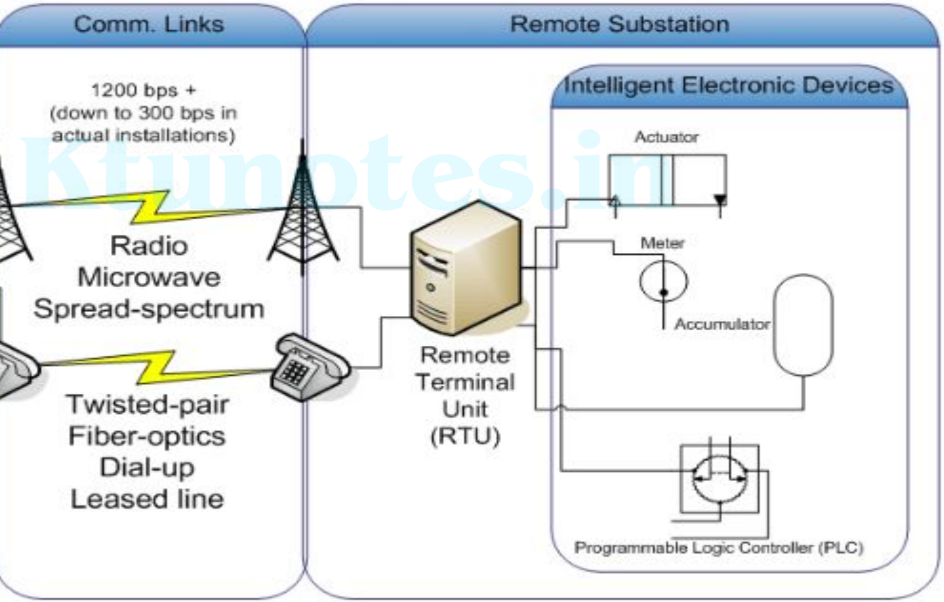
$$\lambda = \frac{I_C}{(1 - \frac{dPL}{dP})}$$

$$0.012P + 8 = 25 \times (1 - 0.2)$$

$$Power\ generation\ P = 1000\ W$$

n be processed by computing machines.

CADA system are mentioned below.



ol

DA refers to sending command messages to a device to operate the and Controls system (I&C) and power-system devices. Conventionally, human managers to initiate command from an operator console on the r. Field personnel can also control machines using front panels.

ollection

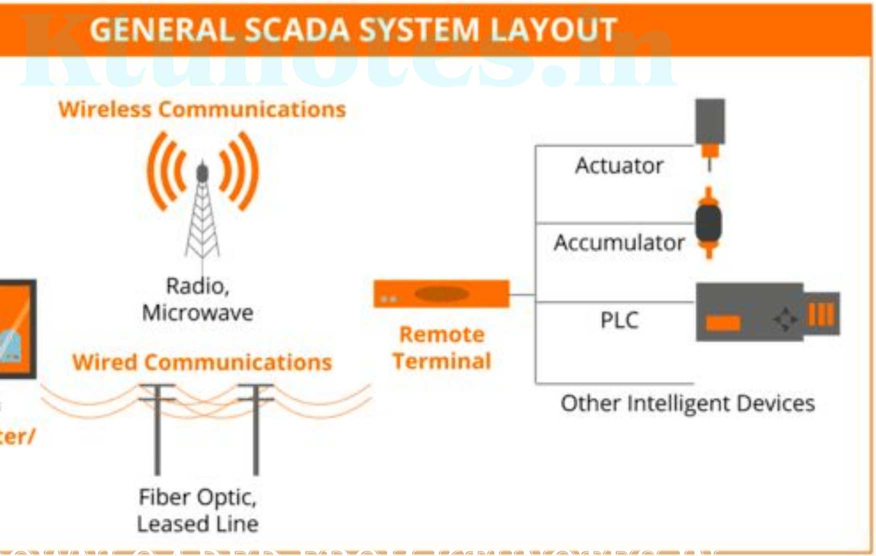
cting data and filling datasheets by hand, SCADA automatically compiles eal-time. SCADA gathers data from hundreds or even thousands of sensors t also generates backlogs for later analysis.

unication:

information to a central hub. A communication network transport all the rom sensors. Earlier systems had radio or modem. Today, SCADA data is internet protocol (IP) and Ethernet.

ents and its functions

ses of the following components



various locations. The more complex a system, the more sensors we may

units:

onsible for collecting data, but we also need something to be able to pret the data. This is where the conversion unit comes in. Conversion units ized units deployed at a specific location in the field. These are connected convert the information they receive into digital format, which is then sent system to display. The two most common types of conversion units used m are PLCs and RTUs. How do we determine which unit we need? that our specification is.

able logic controllers (PLCs)

gic controllers are good for situations where we want more localized rammable logic controller is an industrial digital computer designed for ents and multiple inputs. PLC is used sometimes in place of other due to their versatility, flexibility, affordability, and configuration. However, od programming skills to make the most out of it.

nctions covering a broad area geographically.

units assist as local collection points for gathering information from commands to control and protection relays.

network:

exist without a properly designed communication network system. All ects rely solely on the communication network. It provides a channel etween the supervisory control, the data acquisition units, and any ected to the system. The main function of a communication network n is to connect the Conversion units with the SCADA master station. smitted through various communication platforms such as ethernet, line carrier communication, optical fiber line, cellular, radio, satellite, other wireless protocol. Most facilities have specialized integrated eld buses, wired or wireless, due to security reasons.

considered to be the supervisory computer system because they serve as
centralized processing unit. Although the units themselves are typically
onsoles, there are several other SCADA components, such as software
s, which could be named under this category as well.

g, Human Machine Interface (HMI) is a user interface or dashboard that
n to a machine, system, or device. In a SCADA system, it allows the
and interact with collected and processed data. This interface is usually
tasks like collecting data, creating maps, diagrams, sending out
making reports.

munication server (RCS):

nd analyzing the data gathered in the SCADA system, one needs a digitally
ure place to store this vast database. The human-computer interface (HCI)
y requests data from a server responsible for data acquisition, a
dware that is used to connect software services to the conversion units
e server makes data acquisition from these local units possible.

uninterrupted power supply and a protected environment

ow the status of a complex power system in real-time

onitor and control system that are in remote areas

nsmission and distribution sectors, supervision, monitoring, and ects in all these areas. Therefore, the SCADA implementation of e overall efficiency of the system for optimizing, supervising, and n, transmission & distribution systems. SCADA function in the ffers greater system reliability and stability for integrated grid perations including protection, monitoring, and controlling. To provide minimize operational costs, and to preserve capital investment, the en as seriously in generating stations.

tions of SCADA in power plants include:

nuous monitoring of speed and frequency of electrical machines

aphical monitoring of coal delivery and water treatment processes

city generation operations planning

ol of active and reactive power

and turbine protection and their condition in case of thermal plant

oring of renewable energy farms and load dispatch planning

scheduling

ical data processing of all generation related parameters nce of events recording

e auxiliary systems and supporting components needed to run the nd deliver energy. SCADA system is also highly effective in the e of Plant.

ansmission system:

onding circuit model parameters are often in error as compared he SCADA system. Without a SCADA system, these errors cause the erroneous, and hence, lead to increased costs or incorrect billing. affect state estimator analysis, contingency analysis, short circuit g, machine stability calculations, and transmission planning in case . SCADA integration into the transmission system is significantly relay interface/interaction

gulation management

hanger control

er management

modeling

circuit isolation control and interactive switch control display

eal-time single-line displays

eration and maintenance logs

system diagnostics by using system-defined controller alarms nagement)

CADA implementation to the power distribution not only reduces the
peration and its cost but facilitates smooth processes by reducing

athers the data from various electrical substations and correspondingly in substations continuously monitor the substation components and nsmits that to the central system.

proving efficiency by maintaining a tolerable range of power factor

iting peak power demand

nding and alarming the operators by identifying the problem spot

torian data and viewing that from remote and barely inaccessible locations

ick response to customer service interruptions

der automation and Load Sectionalizer

vide the ability to over-ride automatic control of capacitor banks

ial and abnormal conditions which may affect the quality like
distortions

es of SCADA

tremely advantageous way to run and monitor processes. They are
ons, such as climate control. They can also be effectively used in
s monitoring and controlling a nuclear power plant or mass transit

nance:

e errors by accurately measuring data and increasing the overall

istness:

t of SCADA is performed within a well-established framework that
obustness where power requirement is crucial.

obustness where power requirement is crucial.

ne quality of the produced electric energy profile using standard

and maintenance costs:

are required to monitor field gear in remote locations, this reduces
g costs.

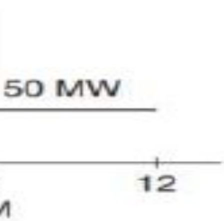
iness systems:

easily integrated with the business systems, leading to increased
ity.

 emented on a large scale in power systems so as to increase their nd durability. Data acquisition and monitoring can be very power systems are upgraded to SCADA. Now, electrical systems l intelligent to monitor and control all of the involved operations become possible only because of technological advancements. s essential for the power sector to optimize their systems as per chnical changes.

$+ 120)10^6$

inimum loads on the units are 100 MW
vely. Determine the minimum cost of
following load (as per Figure given
e cost of fuel is Rs. 2 per million Btu



When the load is 50 MW, for economic loading.

$$\frac{dF_1}{dP_1} = \frac{dF_2}{dP_2}$$

$= 0.04\,P_2 + 3$

$.04P_2 + 1 = 0$

W $P_2 = 41.07$ MW.

on $P_1 = 10$ MW

W, $P_2 = 40$ MW (or)

$P_2 = 50$ MW

, $P_2 = 40$ MW

ion Btu / hr

n Btu / hr

million Btu / hr

$P_2 = 50$ MW

Btu / hr

n Btu / hr

illion Btu / hr

MW, $P_2 = 40$ MW, the fuel cost is less.

$$\frac{}{dP_1} = \frac{}{dP_2}$$

$.04 P_2 + 1 = 0$

0

$_2 = 73.21$

n Btu / hr

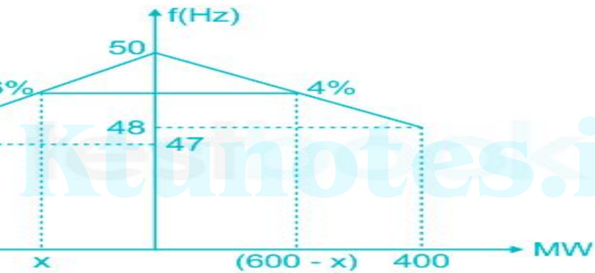
llion Btu / hr

l cost

$2 + 193 + 512.58) (12) (1)$

5

n.



0 MW, speed regulation of 6%

0 MW, speed regulation of 4%

sharing by generator 1

0 MW

generator 2 = 600 − x

rty of triangles

$\frac{00-x}{100}$

generator 2 = 600 − 200 = 400 MW